

Stage 08 — Exploratory Data Analysis (EDA)

EDA is the investigative phase where you **learn the data's story**. It is not a checkbox—its insights shape the entire modeling process.

8.1 Why EDA?

- Reveals distributional shape (skew, tails) that affects model validity.
- Uncovers outliers and data quality issues.
- Surfaces relationships that suggest **feature ideas** and **model forms**.

8.2 A Minimal EDA Checklist

1. Structure & Missingness

- `df.info()`, `df.isna().sum()`, datatype expectations

2. Numeric Profile

- `df.describe()` plus *skew*, *kurtosis*

3. Distributions

- Histograms/KDE + boxplots (outliers)

4. Relationships

- Scatter/line; color by category to reveal clusters

5. Correlation (Carefully)

- Use as a hint; not a causal claim

8.3 Interpretation Patterns

- **Right-skewed targets** → log transforms before linear modeling
- **Outliers** → winsorize or robust estimators
- **Heteroskedasticity** → consider variance-stabilizing transforms
- **Seasonality/trends** → calendar/time features later

8.4 Communication

- Every plot gets a plain-English caption: *What? So what? Now what?*
- Conclude with **3 insights + assumptions** to guide preprocessing & feature engineering.

8.5 Example Diagram (suggested slide)

Raw Data → Profiling → Visual Patterns → Insights → Assumptions → Next Steps

(Cleaning/Features)